

Poisoning-Assisted Membership Inference in Federated Learning

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Abstract—Federated learning (FL) is a distributed machine learning approach that allows multiple clients to collaboratively train a global model without sharing their local raw data. Despite this advantage, FL remains vulnerable to membership inference attacks (MIAs), which can compromise the membership privacy of federated clients. Nevertheless, existing MIAs exhibit significant performance degradation when the malicious clients lack knowledge of the target model and the training data of benign clients. In this paper, we propose a novel scheme, Temporal Evolution of the Poisoning Effects (TEPE), to conduct MIAs against black-box FL models without any prior knowledge of the target models or benign clients' training data. Specifically, we design a two-stage inference method: the poisoning process and the scoring process. For the poisoning process, we propose three poisoning strategies for three different auditing requirements. For the scoring process, we design a novel quantification method and an unsupervised inference model to extract membership features within the effects of poisoning attacks. The proposed attack leverages the intuition that if a sample is used by FL benign clients, its prediction may not be readily altered by poisoning attacks. Finally, we propose a modified random response algorithm (RR-Group) to detect our attacks, which can guarantee the performance of FL models and effectively reduce the attack performance of TEPE. Extensive experiments demonstrate that TEPE achieves competitive inference accuracy and F1-score compared to the most MIAs, while evading two representative defenses, except for our proposed detection method RR-Group.

Index Terms—Federated learning, membership inference attack, membership inference defense, poisoning attack, temporal evolution.

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I. INTRODUCTION

AS A distributed machine learning paradigm, federated learning (FL) [1], [2], allows clients to collaboratively train a global model on their local datasets without exposing or transferring their private raw data [3], [4]. Some works [5], [6], [7] show that, by encapsulating local updates, local training algorithms, and FL model parameters in a hardware-backed trusted execution environment (TEE) [8], [9], FL clients can prevent adversaries (in black-box settings) from accessing sensitive data. FL is widely used in many real-world applications, such as content generation [10], smart city [11], financial system [3], and disease prediction [12]. However, recent works reveal that FL is confronted with several privacy issues, such as source inference attacks [13], inversion attacks [14], category inference attacks [15], property inference attacks [16], and membership inference attacks (MIAs) [17], [18], [19], [20].

MIAs threaten the privacy of FL models, as they can infer whether a sample has been used in a given FL model. For example, when an FL model is trained on datasets from multiple medical institutions, once adversaries confirm that a sample participates in the training process, they may indirectly reveal sensitive information such as disease history and treatment records of patients [21]. Therefore, understanding and defending against MIAs is critical to ensure the privacy of FL models. However, MIAs against FL still face several challenges. First, the discriminative features between FL clients' member and non-member samples typically exhibit minimal divergence, rendering them challenging to distinguish effectively. Second, membership information becomes more concealed due to FL's aggregation mechanisms, resulting in lower attack performance. Third, extensive membership inference defenses in FL significantly constrain adversaries' capabilities, thus making the MIAs' intermediate process difficult to implement. These inherent limitations consequently lead to suboptimal attack performance.

Most of the existing MIAs build upon common assumptions that adversaries can obtain some prior knowledge about either the target models (such as model structure¹ [17], [22], [23], parameters [17], [24], [25], or training loss function [26]) or benign clients' training data (such as data samples [22], [27] or data distribution [21], [27], [28], [29], [30]), as shown in

¹MIAs designed for a particular model are considered to possess prior knowledge of model structure.

TABLE I
ATTACK METHODS AND THEIR PRIOR KNOWLEDGE REQUIREMENTS OF
TARGET MODEL AND TRAINING DATA

Attack	Target Model			Training Data	
	Structure	Param	Loss	Sample	Distribution
White-Box [17]	✓	✓	×	×	×
Shadow [22]	✓	×	×	✓	✓
Label-Only [23]	✓	×	×	×	✓
CMI [24]	✓	✓	×	✓	×
Yuan [25]	✓	✓	×	✓	×
Trajectory [26]	✓	×	✓	×	✓
TEAR [27]	×	×	×	✓	✓
Song [21]	✓	×	×	✓	✓
Tramèr [28]	✓	×	×	×	✓
Chen [29]	✓	×	×	×	✓
L-Leaks [30]	×	×	×	×	✓
Truex [37]	×	×	×	✓	×
AggrEvader [38]	✓	×	×	✓	✓
Mo [39]	✓	✓	✓	×	×
Pichler [40]	✓	✓	×	×	×
FedMIA [41]	✓	✓	✓	×	✓
TEPE (our)	×	×	×	×	×

Required Knowledge: ✓ Non-required Knowledge: ×

Table I. However, existing MIAs face inherent limitations in FL due to client-side safeguards. Federated Learning as a Service (FLaaS) [31], such as BigDL-PPML [32] and YiduManda [33] can deploy their FL models as black-box models using TEE with Software Guard Extensions [34]: all FL clients can only access the global model through the provided black-box APIs, where FL clients can only control their own local data [7]. Consequently, under these constraints, existing MIAs are unable to utilize the prior knowledge to facilitate the execution of the attack process, thereby leading to a decline in attack performance.

In this paper, we propose a Temporal Evolution of Poisoning Effects (TEPE) scheme to ensure that MIAs are executed with mere black-box access by constructing targeted poisoning attacks. Our intuition is that the samples that have not been trained by FL clients are more vulnerable to poisoning attacks [35]. The underlying rationale is that training samples often exhibit a contradictory gradient [36] with poisoned samples during gradient updates, whereas non-training samples do not. By detecting this difference, we can implement a successful MIA, which consists of three major parts. First, three targeted poisoning strategies are developed to cooperate with the leakage of membership information of target samples. Since poisoned samples generated from target samples can reduce the performance of the target samples in the model, the effects of member samples and non-member samples may be opposite. Second, we propose a scoring method to quantify poisoning effects from the FL training process. Due to the uncertainty of the model and training process, the temporal evolution of poisoning effects is usually complex and unordered. Our designed unsupervised inference model learns the characteristics of temporal evolution. Third, a membership inference defense is proposed to protect the membership information of FL clients under TEPE. We evaluate TEPE's excellent attack performance under some membership inference defenses. The presented defense method considers the privacy leakage under targeted poisoning attacks, which makes it effectively balance model performance and privacy protection.

As a result, FL clients with the inference defense method can precisely defend against poisoning MIAs.

The contributions of this paper are summarized as follows.

- We propose a membership inference attack scheme, TEPE, against black-box FL models. TEPE does not require any prior knowledge about the target models or benign clients' training data but only obtains the global model's prediction vector.
- We propose three poisoning strategies against three conditions. Then we propose a scoring method, which is the first time to show how the temporal evolution of poisoning effects can be utilized as a membership feature against FL models.
- We propose a membership inference defense scheme, RR-Group, based on Random Response, to balance model performance and privacy preservation against TEPE.
- Extensive experiments on different models and datasets show that TEPE outperforms the mainstream methods in attack performance, and RR-Group can provide better protection than two representative defenses.

The remainder of this paper is organized as follows. Section II reviews existing work relevant to this paper. Section III introduces the threat model. Section IV gives the detailed design of TEPE. Section V presents the defense method. Section VI presents the performance evaluation of TEPE, followed by the conclusion of this paper in Section VII. The code for TEPE is released for the purpose of reproducibility.²

II. RELATED WORK

A. Membership Inference Attacks Against Federated Learning Models

Some researchers have employed shadow training techniques [22] to investigate the risk of membership privacy leakage from the training data. Truex et al. [37] use shadow datasets and shadow models to simulate the behaviors of target models and build inference models to achieve MIAs. Since FL models require the exchange of parameter updates during the training process, it provides new avenues for MIAs. Nasr et al. [17] develop MIA strategies, both supervised and unsupervised, by extracting attribution information of specific samples from observed model parameters updates. The key idea behind TEAR [27] is to quantify the adversarial robustness of target samples using adversarial examples and utilize this information to train a supervised inference model that distinguishes between the member and non-member samples. As the logits of the target models can provide more predictive information, Yan et al. [30] propose L-Leaks, which can allow adversaries to utilize the distribution information of training data from the substitute models to perform MIAs. Yuan et al. [25] analyze the membership of the federated recommendation system to perform MIAs by observing the parameter updates of local models. The strategy proposed by Pichler et al. [40] is applicable to models with ReLU activations and leverages the specific properties of this activation function to achieve perfect

²Our code is available at <https://github.com/Zili1000/TEPE>

accuracy. FedMIA [41] significantly enhances the attack efficacy by introducing a one-tailed likelihood ratio hypothesis test based on non-target client updates and leveraging an “all for one” principle to aggregate update information from all clients across multiple communication rounds.

However, MIAs in FL typically require adversaries to access parameter updates, modify the training process, or possess some prior knowledge of training samples. In practice, model parameters and gradient information are often encrypted, and the FL training process may be conducted within a hardware-backed TEE. Due to these limitations, most existing MIAs struggle to work in the black-box FL model scenario. Different from the existing solutions, we consider an extremely constrained scenario and propose a practical MIA that requires only the black-box API.

B. Poisoning Attacks for Privacy

Recently, some works focus on poisoning attacks [42], [43], [44], [45], and some researchers have noted that poisoning attacks can pose a threat to the privacy of models. Tramèr et al. [28] demonstrate that if adversaries can poison training sets, the model trained on the poisoned dataset can leak important private details using only a small number of poisoning samples. Chen et al. [29] propose that poisoning attacks can expose the membership of target classes. They can improve the accuracy of MIAs with dirty-label and clean-label poisoning attacks. He et al. [46] utilize poisoning attacks and sequence prediction confidence to implement MIAs by shadow models. AgrEvader [38] implements MIAs by adopting a mechanism of “covering” local updates trained on clean samples, maximizing adversarial effects on target samples while evading detection by Byzantine-robust mechanisms. ARO [47] actively induces robust overfitting during pre-training. This strategy amplifies membership leakage in downstream tasks without compromising the model’s primary accuracy, while simultaneously maintaining the stealthiness of the attack.

However, the above works demonstrate that poisoning attacks can only expand the privacy leakage risks of models, instead of leveraging the effects of poisoning attacks to violate model privacy. While they combine data integrity and confidentiality, the direct purpose of compromising data integrity is not to violate confidentiality but to study the effects of integrity on confidentiality. Our scheme to execute MIAs through poisoning effects further delves into the relationship between membership inference attacks and poisoning attacks.

C. Membership Inference Defense

Some recent works [48], [49] focus on membership inference defenses. Considering the defense stage, the existing defenses can be roughly divided into two categories: inference-stage defenses and training-stage defenses. For inference-stage defenses, MemGuard [50] is a classic defense method, which confounds membership information by adding adversarial noise to the model output. For training-stage defenses, LoDen [51] enhances the robustness of FL against MIAs by enabling clients to filter out vulnerable training samples collaboratively, including those

under MIAs and their highly similar counterparts, thereby reducing the FL model’s exposure to adversaries. Zheng et al. [24] suggest adopting an improved random response (RR) [52], namely RR-Label, to reduce the accuracy of membership inference, thereby solving the label bias issue under non-i.i.d. In addition, Dropout [53], differential privacy (DP) [54] and regularization [55] can also be used to defend against MIAs. DP-SGD [56] adds Gaussian noise to the gradients of the target model during the training process. Nasr et al. [57] propose that the adversarial regularization can significantly generalize the model and guarantee membership privacy. MemDefense [58] safeguards membership privacy by deploying two key components—a parameter filter and a noise generator—in each federated learning round. These components collaboratively inject carefully crafted pruning perturbations into the local models. Hu et al. [59] propose PDP-Flames, a novel differentially private framework that capitalizes on the sparse gradient of federated knowledge graph embedding. By integrating advanced private selection techniques, it achieves a superior privacy-utility trade-off.

Although the existing membership inference defenses can protect privacy in the absence of poisoning attacks, they struggle to achieve an effective balance between model performance and privacy protection, particularly under targeted poisoning attacks (e.g., TEPE). In such cases, conventional defenses may fail, as the privacy leakage induced by poisoning attacks can significantly outweigh the perturbations introduced by defenses. Therefore, we propose a defense algorithm, whose key advantage lies in its capability to maintain the optimal balance between model performance and privacy preservation, even under targeted poisoning attacks.

III. THREAT MODEL

A. Target Model

We focus on exploring the membership information in FL black-box models. FL clients, including malicious clients and benign clients, can access the prediction interface of the global model and query the global model $\mathcal{M}$ using their data to obtain the corresponding prediction results. In the FL setting, the considered target model is a combination of a series of global models $[\mathcal{M}^1, \mathcal{M}^2, \dots, \mathcal{M}^T]$, where T represents the number of the FL training iterations.

B. Malicious Clients’ Knowledge

As mentioned in [5], [6], [8], [32], [60], the local training process occurs within a black-box setting and FL model parameters are encrypted in practice. Therefore, we consider weak but more practical malicious clients that only have the prediction results of the global models and do not possess any prior knowledge of the following information:

- *The structure of the target model:* including the type and structure of the target model, and the lack of this information makes the shadow models impossible to build.
- *The parameters of the target model:* including the internal parameters, the regularization parameters, and the other hyperparameters.

- *The global training process of FL*: including the global training process and the hyperparameter settings controlled by the FL central server, such as client weights and the global learning rate.³
- *Auxiliary datasets or training data distribution*: including any dataset that is distributed similarly to the training set utilized to create the target model, as well as the distribution of the training data.
- *Membership information of other clients' training data*: including any samples from other benign clients, and malicious clients do not have any information about the membership attributes of each sample. Specifically, malicious clients cannot utilize known membership information to build a supervised inference model.

C. Malicious Clients' Capability

Malicious clients can collect their own dataset D^{mal} without prior knowledge of other clients' training data, such as web crawling, artificial construction, and interview surveys. The malicious clients can adaptively change their poisoning strategy during the training process. Their datasets are independent of other benign clients' datasets. The data held by FL clients could potentially undergo scrutiny from trusted third parties. For instance, a hospital's data sources might be subject to verification by an oversight health authority. We set three audit scenarios: none, full, and partial (corresponding to the three poisoning strategies in Section IV-A). In each iteration, malicious clients use their datasets to train the global model and upload the model to the server⁴. They can only access the global model's APIs using the sample $z = (x, y)$, where x is the input feature, y is the truth label. For neural networks, the output of the second-to-last layer is logits, which we represent as the vector $g_M(z)$. The softmax function is then computed on the logits to obtain the final prediction vector. The predicted value corresponding to category ι is calculated as

$$F_M(z)_\iota = \frac{\exp(g_M(z)_\iota)}{\sum_{\varsigma=0}^{C-1} \exp(g_M(z)_\varsigma)}. \quad (1)$$

$F_M: R^d \rightarrow R^C$ is the interface function of an FL global model, with d input features and C output classes. For the sample $z = (x, y)$, the model outputs a prediction vector $F_M(x)$ covering all class labels, with $\sum_{\varsigma=0}^{C-1} F_M(x)_\varsigma = 1$. For a training set D , the model training process is to minimize the prediction loss across all training samples, and we have

$$\min_{\theta} \frac{1}{|D|} \sum_{z \in D} \mathcal{L}_M(z), \quad (2)$$

where $|D|$ represents the size of the client's training set, $\mathcal{L}(\cdot)$ represents the loss function, and θ represents model internal parameters. The key symbols are detailed in Table II.

³The above settings are the same as [27], [30].

⁴All client-side operations, including model download, local training, and model upload, are confined to a hardware-based TEE [5], [6], [8]. This ensures the confidentiality and integrity of the FL process [32], [33], making it completely opaque to the outside world.

TABLE II
DESCRIPTION OF KEY SYMBOLS

Symbol	Description
$\mathcal{M}$	model
θ	the model internal parameters
$\mathcal{T}$	the total number of the FL training iterations
n	the number of FL clients aggregated
z	sample
z_t	target sample
z_{mal}	poisoned sample (z_{adv}, z_{spy})
x	the sample's input feature
y	the sample's truth label
y_p	the sample's poisoned label
F_M^x	the prediction result
C	the number of categories in the dataset
D	dataset
D^{mal}	malicious client's dataset (D^{adv}, D^{spy})
D'	inference dataset
$\mathcal{A}$	inference model
α	Spy's poisoned sample number
β	Adv's poisoned sample number
$\mathcal{N}$	initial Group's size

D. Malicious Clients' Goal

Given a target sample $z_t = (x_t, y)$ and a series of global models $[\mathcal{M}^1, \mathcal{M}^2, \dots, \mathcal{M}^T]$, malicious clients' goal is to determine whether z_t exists in the training sets of other clients, and we have

$$\mathcal{A}(\text{Score}(z_t | [\mathcal{M}^1, \mathcal{M}^2, \dots, \mathcal{M}^T])) \rightarrow \text{In/Out}, \quad (3)$$

where $\mathcal{A}(\cdot)$ represents the inference model of TEPE, $\text{Score}(\cdot)$ serves as the foundation for evaluating the membership information of z_t and constitutes the core component of TEPE, and the label *In* (resp. *Out*) indicates that z_t belongs to the target model's training set (resp. testing set).

IV. DESIGN OF TEPE

To assess the membership privacy of an FL model, we design an MIA dubbed TEPE from the perspective of FL clients. TEPE utilizes the FL training process and leverages the difference in the temporal evolution of the poisoning effects on the member and non-member samples. If a sample z_t is a member (i.e., it is part of the FL training set), the model parameters θ already reside near a local minimum of its loss function $\mathcal{L}_M(z_t)$. In this case, the poisoning update direction may oppose or form a significant angle, thereby exerting a relatively minor impact on the prediction accuracy for z_t . Conversely, for non-member samples, whose loss landscape has not been optimized by the model, the same poisoning update can alter the FL model's prediction behavior more effectively. As shown in Fig. 1, TEPE mainly consists of two components:

Poisoning Process: We propose three poisoning strategies to generate a series of poisoned samples using the target samples and put them into the FL training process. Assuming that with the targeted poisoning attacks, the poisoned label y_p is different from the truth label y . We define three distinct data poisoning scenarios:

- *Adv*: The malicious clients have complete control over their local data environment, allowing them to freely select or

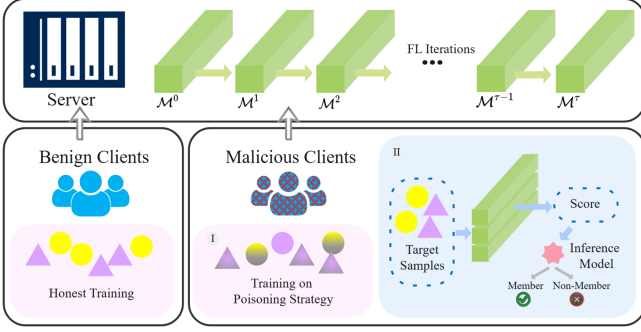


Fig. 1. Overview of TEPE against FL models. In each iteration, benign clients participate in the honest training process, whereas malicious clients generate poisoned samples based on their specific poisoning strategies for local training. Subsequently, these malicious clients may perform membership inference by analyzing the global model's outputs through a scoring method to construct an inference model. Crucially, from the server's perspective, malicious and benign clients are indistinguishable; both download the global models, perform training on their respective datasets, and upload the updated model parameters. Scoring process is secretly initiated by the malicious clients, as shown in Fig. 2.

craft any poisoned samples without fear of external audit or interference. This scenario is commonly encountered in personal endpoint applications, such as keyboard input prediction [61], voice assistants [62], and personalized recommendation systems [63].

- *Spy*: The malicious clients operate under scrutiny, where a trusted third party (e.g., auditors or escrow agents) may review the local dataset. Poisoned samples must be crafted to evade detection, mimicking benign data to avoid being excluded. This configuration is frequently observed in financial scenarios [3], where institutions such as banks and insurance companies are typically subject to stringent regulatory scrutiny.
- *Adv&Spy*: The hybrid scenarios are more complex and diverse, such as smart cities [11] and industrial Internet of Things [64] under various auditing requirements. FL clients may be subject to varying audit requirements due to factors such as differing confidentiality clearance levels and heterogeneous hardware conditions. It is therefore reasonable to expect that some clients may undergo key audits, while others may be prohibited from audit or lack the capability.

Scoring Process: We propose a scoring method to quantify poisoning effects on prediction results $F_{\mathcal{M}^T}(z)$ during each iteration of the FL training process. Malicious clients can score the target samples on the series of models $[\mathcal{M}^1, \mathcal{M}^2, \dots, \mathcal{M}^T]$ to obtain the evolution of scores as the attack dataset. Then, we propose a modified unsupervised model as the inference model to extract membership information from the attack dataset (as shown in Fig. 2).

A. Poisoning Process

Building effective poisoning strategies to compromise membership privacy against FL poses two challenges. First, existing poisoning attacks [36], [65], [66] are primarily designed to degrade model accuracy of a particular class rather than to breach

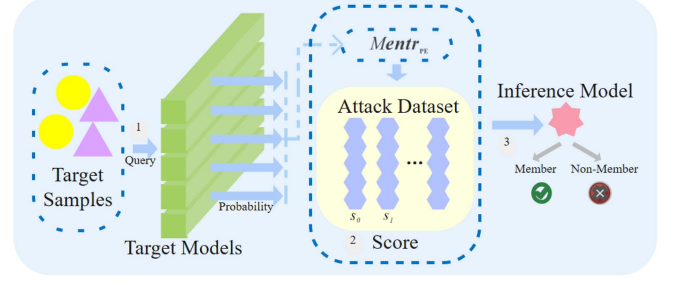


Fig. 2. Scoring process. The malicious clients initiate the attack by querying the target models (i.e., the global model from each training iteration) with the target samples to obtain their probabilities. Subsequently, these probabilities are scored using (8) and (10) to construct an attack dataset. Finally, a binary unsupervised inference model is built with this dataset to execute the membership inference attack.

a specific sample's privacy, as follows:

$$\mathcal{M}_p = \min_{\theta} \frac{1}{|D^{mal}|} \sum_{z_a} \mathcal{L}_{\mathcal{M}^T}(z_{mal}), \quad (4)$$

where $\mathcal{M}_p$ represents the poisoned model and z_{mal} indicates the poisoned sample belonging to the malicious client's dataset D^{mal} . The malicious aggregation process is as follows:

$$\mathcal{M}_g = \frac{1}{n} \sum_{\varsigma=1}^{n-1} \mathcal{M}_{\varsigma} + \frac{1}{n} \mathcal{M}_p, \quad (5)$$

where $\mathcal{M}_g$ represents the global model and n indicates the number of clients aggregated. The malicious client's generated poisoned sample $z_{mal} = (\hat{x}, \hat{y}_p)$, where $\hat{y}_p \neq y$ and $\hat{x} \neq x_t$, cannot determine which samples may be affected, failing to infringe upon the membership privacy of the target sample. Second, due to the FL aggregation mechanism, the previous poisoning effect becomes even less ideal with interference from other benign clients, further hiding the membership information within the global model.

To obtain the distinguishable information, we propose three targeted poisoning strategies that interfere with the decision boundary to counteract the FL aggregation effect and associate the effects of poisoning with target samples.

1) *Adv*: The FL clients are not audited in this setting, and the malicious clients can freely set their datasets. We consider a simple strategy, where the malicious clients' poisoned samples $z_{adv} = (x_t, y_p)$ adopt the dirty-label poisoning attack [65], keeping the input features unchanged and replacing the labels with poisoned labels. In practical scenarios characterized by high-dimensional and sparse feature spaces, as well as non-i.i.d. data distributions, the presence of a very small number of instances with identical features but differing labels in an FL client's dataset can easily evade the server's anomaly detection mechanisms. This inherent stealth enables adversaries to deploy such attacks effectively in real-world settings.

2) *Spy*: We consider a more challenging setting wherein each FL client's dataset is subject to audit by the trusted third party. As a result, poisoned samples in the Adv's setting (original features and abnormal labels) can be visually identified and readily

removed from the dataset prior to training. Since the malicious clients can only obtain the prediction vector without any prior knowledge about the model or training data, we consider a poisoning strategy, which aims to generate poisoned samples $z_{spy} = (x^*, y_p)$ to disturb the decision boundary around the target sample. x^* denotes the input feature of the poisoned sample, generated through an optimization process. Our optimization method calculates the L2 norm of the model prediction output for the initialized sample and the target sample x_t , and formulates the objective function as

$$x^* = \underset{x_0}{\operatorname{argmin}} \|F_{\mathcal{M}^I}(x_t) - F_{\mathcal{M}^I}(x_0)\|_2, \quad (6)$$

where $\|\cdot\|_2$ represents the L2 norm of a vector. We optimize (6) using the Adam [67] optimizer to refine the initialized noise sample continuously. The poisoned samples with similar features gradually move closer to the target samples in the feature space of the global model, achieving the effect of disturbing the decision boundary around the target samples (as shown in Fig. 6). To further reduce the possibility of being audited, the poisoned labels are set as similar as possible to the truth labels (e.g., labeling a “shrew” as a “mouse”), and we discuss the poisoned label selection in VI-B2. With an average production time of merely 15 seconds per poisoned sample and a minimal required proportion of such samples, the associated computational overhead is negligible. This demonstrates the practicality of our attack, even within large-scale federated learning systems.

3) *Adv&Spy*: In the hybrid setting, we consider the collaborative strategy against the target samples by the malicious clients operating under different audit conditions. The malicious clients subject to strict auditing execute the Spy strategy, while those not under scrutiny carry out the Adv strategy. The potential for collaboration among multiple malicious clients aligns with real-world requirements, as in complex scenarios involving hundreds of clients, multiple malicious clients can more easily blend into the system and evade detection.

The Adv strategy and Spy strategy induce the target model to mark the target samples as outliers, thereby increasing the model’s attention toward them, which highlights the difference between member and non-member samples. To increase the malicious clients’ impact on the model, we duplicate the poisoned samples to enhance the attack effectiveness.

B. Scoring Process

1) *Quantifying Poisoning Effects*: During the normal training process, the predicted value of the truth label continuously increases, while the predicted values of other labels continuously decrease. Suppose a malicious client introduces a poisoned dataset as its own training set and participates in the FL model training process. In that case, the training algorithm similarly should minimize the prediction loss of the poisoned samples, which causes the model’s predictions to shift toward the poisoned label. Consequently, the predicted value of the poisoned label may ascend, while that of the truth label may descend.

To quantify the effects of the poisoning attacks, we design a scoring method that possesses the following two properties:

Algorithm 1: Poisoning Effects Quantification Algorithm.

Input: Target sample $z_t = (x_t, y)$, the $\mathcal{I}$ th iteration target model $\mathcal{M}^I$, poisoned label y_p , Adam’s max iteration number γ and learning rate η , Spy’s poisoned sample number α , Adv’s poisoned sample number β

Output: Membership metric $Mentr_{PE}^I$

```

1: Spy:
2: for  $\nu = 1$  to  $\alpha$  do
3:    $x_0 \leftarrow$  Initialize  $x_t$  with random noise
4:    $x^* \leftarrow$  Adam( $\|F_{\mathcal{M}^I}(x_t) - F_{\mathcal{M}^I}(x_0)\|_2, \eta, \gamma$ )
5:   Spy’s training set  $D^{spy}$  add  $z_{spy} \leftarrow (x^*, y_p)$ 
6: end for
7:  $\theta_{spy} \leftarrow$  Train with  $D^{spy}$ 
8: Adv:
9: for  $\nu = 1$  to  $\beta$  do
10:  Adv’s training set  $D^{adv}$  add  $z_{adv} \leftarrow (x_t, y_p)$ 
11: end for
12:  $\theta_{adv} \leftarrow$  Train with  $D^{adv}$ 
13:  $F_{\mathcal{M}^I}(x_t)_y, F_{\mathcal{M}^I}(x_t)_{y_p} \leftarrow F_{\mathcal{M}^I}(x_t)$ 
14: return  $Mentr_{PE}^I(F_{\mathcal{M}^I}(x_t)_y, F_{\mathcal{M}^I}(x_t)_{y_p})$ 

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- *Property 1*: For a truth label, the metric value is supposed to decrease monotonically in tandem with the rise in the truth label’s prediction probability. In other words, the truth label’s prediction is associated with a lower metric value, signifying a less pronounced impact on poisoning.
- *Property 2*: For a poisoned label, the metric value should monotonically increase as the poisoned label’s prediction probability increases. This means that the higher poisoned label’s prediction probability corresponds to a larger metric value, indicating a stronger impact on poisoning.

Some works [22], [68] based on entropy explain the existence of privacy risks by leveraging the difference in entropy distribution between the training data and testing data. They use Cross-Entropy [69] as the measure of entropy, and it satisfies

$$-\sum_{\iota} F_{\mathcal{M}^I}(x)_{\iota} \log(F_{\mathcal{M}^I}(x)_{\iota}) \leq \hat{\tau}_y, \quad (7)$$

where $\hat{\tau}_y$ is a threshold using shadow training techniques. If the predictive entropy is less than the preset threshold, the malicious clients classify the sample as a member sample; otherwise, it is classified as a non-member sample. However, (7) cannot contain any information about the labels and cannot reflect the changes in different categories during the training process, ignoring a lot of information about privacy leaks. To address this challenge, inspired by Focal loss [70], we propose a scoring method for the predicted values of the truth labels and the poisoned labels, as shown in Algorithm 1, and we have

$$\begin{aligned}
Mentr_{PE} = & -(1 - F_{\mathcal{M}^I}(x)_y) \log(F_{\mathcal{M}^I}(x)_y) \\
& - F_{\mathcal{M}^I}(x)_{y_p} \log(1 - F_{\mathcal{M}^I}(x)_{y_p}).
\end{aligned} \quad (8)$$

Interpretation: Focal loss is expressed as

$$FLoss(p) = \begin{cases} -(1-p) \log p & \text{if positive,} \\ -p \log (1-p) & \text{if negative,} \end{cases} \quad (9)$$

where p is the predicted probability. Focal loss is initially designed to address the issue of imbalance between positive and negative samples. In the training set, easy-to-classify samples typically far outnumber hard-to-classify ones. Given this circumstance, the core concept of Focal loss lies in modifying the overall loss, which augments the weight of those hard-to-classify samples and simultaneously diminishes the weight of easy-to-classify samples. Focal loss allows the model to focus more on hard-to-classify samples during the training process. $-(1-p) \log p$ is a monotonically decreasing function satisfying Property 1, and $-p \log (1-p)$ is a monotonically increasing function satisfying Property 2. We have two reasons: i) For the poisoned label, since poisoned samples inherently constitute a small portion of the total training set, the likelihood of a target sample z_t being predicted as the poisoned label during normal training is very low, and the poisoned label should be classified as the hard-to-classify class. ii) For the truth label, the model already has a large number of correctly labeled training samples, making it easier for the target sample to obtain a higher value on the truth label, and the truth label should be classified as the easy-to-classify class. $Mentr_{PE}$ can be interpreted as a variant of conditional entropy, operating between the poisoned labels and the true labels. From an information-theoretic perspective, $Mentr_{PE}$ quantifies the uncertainty or information content associated with the joint event where, under poisoning interference, the model's confidence in a true label y , denoted as $F_{M^T}(x)_y$, decreases while its confidence in the poisoning label y_p , $F_{M^T}(x)_{y_p}$, increases. For non-member samples, whose model parameters have not been optimized on the loss landscape of x_t , the poisoning attack can induce this confidence flip more effectively, resulting in a higher $Mentr_{PE}$ value. From the viewpoint of the loss function, the two components of $Mentr_{PE}$ correspond to distinct learning pressures. $-(1 - F_{M^T}(x)_y) \log(F_{M^T}(x)_y)$: This represents a *positive incentive* for the true label. $-F_{M^T}(x)_{y_p} \log(1 - F_{M^T}(x)_{y_p})$: This acts as a *negative incentive* for the poisoned label. Consequently, $Mentr_{PE}$ adeptly captures the bidirectional effect induced by a poisoning attack on a specific sample. This discrepancy forms the foundation of our membership inference.

We construct a scoring evolution vector using a series of metrics on the global models for a target sample, and we have

$$s = [Mentr_{PE}^1, Mentr_{PE}^2, \dots, Mentr_{PE}^T]. \quad (10)$$

Fig. 3 shows the evolution of metric values of the poisoning effects on target samples. As the FL training progresses, non-member samples' metric value should gradually increase, while member samples' metric value, influenced by original training samples, should not follow the same temporal evolution.

2) *Inference Model Construction:* Although a preset threshold needs to be set to distinguish samples in [21], these temporal evolution poisoning effects for FL models can distinguish the difference between the member and non-member samples without thresholds. Therefore, we propose an unsupervised inference

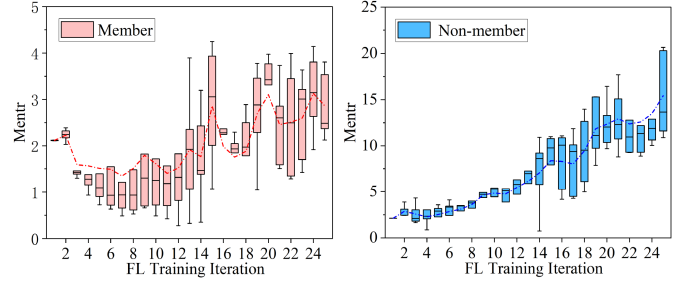


Fig. 3. Key observations on the temporal evolution of poisoning effects under the FL training iterations. The setup remains the same as the CIFAR-10 experiment in Section VI. It can be observed that as the FL training process proceeds, the poisoning effects on both the member (left) and non-member (right) samples exhibit distinct temporal evolution trends between the two sets. This disparity implies that the temporal evolution of poisoning can serve as a viable indicator for distinguishing between the member and non-member samples.

model, which is a binary classifier with *In* and *Out* classes. Regarding the clustering, we simply perform the K-means [71] procedure to obtain cluster assignments and no longer discuss other clustering methods.

For each target sample, we calculate s using (10) to form the unsupervised inference dataset D' . Considering that the evolution of the metrics also contains important membership information, we make improvements to the calculation of the distance between vectors of K-means to better align with the data characteristics of the metrics. We use the distance between slopes to reflect this characteristic. The modified distance equation is expressed as

$$dist(s_{ij}, c_{kj}) = (deriv(s_{ij}) - deriv(c_{kj}))^2 + (s_{ij} - c_{kj})^2, \quad (11)$$

$$deriv(v_{ij}) = \frac{v_{i(j+h)} - v_{i(j-h)}}{2h}, \quad (12)$$

where c_k is the k -th cluster center, s_{ij} and c_{ij} are the values of s_i and c_k in the j th dimension, respectively, $deriv(\cdot)$ is used to calculate the slope of a vector. We use the central difference method to estimate the derivative of the vector and h is the offset distance forward and backward to calculate the slope of the current dimension.

Due to the sensitivity of D' to temporal evolutions, we adopt Dynamic Time Warping (DTW) [72] to facilitate our unsupervised clustering analysis. It is a strategy based on dynamic programming that aligns two time series nonlinearly through timing alignment to correctly calculate the similarity between them, and we have

$$DTW(s_{ij}, c_{kj}) = \min(DTW(s_{i(j-1)}, c_{k(j-1)}), DTW(s_{i(j-1)}, c_{k(j)}), DTW(s_i, c_{k(j-1)})) + dist(s_{ij}, c_{kj}), \quad (13)$$

where DTW is a $j \times j$ matrix initialized to infinity and then specific values are calculated step by step. Then, we use the square root of $DTW(s_{ij}, c_{jk})$ as the basis for clustering, and

we have

$$Dist(s_{ij}, c_{jk}) = \sqrt{DTW(s_{ij}, c_{jk})}. \quad (14)$$

Each sample is assigned to the nearest cluster center by calculating its distance to the cluster center until the cluster center converges. We label the cluster with the larger metric values *Out* and the other *In*.

V. POTENTIAL DEFENSES

In FL, malicious clients can launch MIAs to violate the privacy of benign clients. Many works, as described in Section II-C, are proposed to defend against the leakage of membership privacy. Next, we analyze the limitations of existing defenses against TEPE and propose our defense method.

A. Inference Stage Defenses

Confidence score masking is mainly used for defense in black-box scenarios and provides a guarantee for membership privacy. MemGuard [50] aims to change the true confidence scores returned by the target model during the inference stage, thereby blurring the difference between the member and non-member samples. However, FL malicious clients do not need to take any perturbation measures to reduce the attack accuracy. When malicious clients can directly access the posterior output of the model, MemGuard is ineffective. Even if malicious clients are constrained to use MemGuard to perturb the confidence scores, MemGuard's $L2$ retains the possibility of privacy leakage as

$$L2 = \text{ReLU}(-F_{\mathcal{M}^T}(x)_c - e + \max_{i \neq c} \{F_{\mathcal{M}^T}(x)_i + e\}), \quad (15)$$

where e is the noise. (15) ensures that the label corresponding to the highest prediction confidence remains unchanged, which may preserve the magnitude of the relationship between the poisoned labels and the truth labels in TEPE.

B. Training Stage Defenses

1) *Disturbance Gradient*: DP-SGD [56] achieves ϵ -DP by adding noise to the model training gradient. TEPE may be affected by DP-SGD because of the influence of the poisoned samples with the noise on gradients during the FL training process, and be misled to distinguish member and non-member samples.

2) *Disturbance Training Set*: Zheng et al. [24] suggest using a random response method called RR-Label to defend against membership inference attacks by perturbing the labels of the training set. RR-Label expands the alternative label of RR-Topk [52] to the remaining labels, thereby addressing the label bias issue. However, the random response results are influenced by targeted poisoning strategies in Section IV-A. For example, suppose a client has some protected data from C_1 classes. Under the influence of malicious clients, the model's prediction for the target sample may be the other $C - C_1$ classes, which leads to RR-Label still retaining the original label of the protected data. We denote the other classes as $C \setminus C_1$. Even if malicious clients are unaware of $C \setminus C_1$, (8) quantifies the impact of poisoning attacks only on the prediction values of the poisoned labels

Algorithm 2: RR-Group.

Input: Benign client's training set D^{ben} , the $\mathcal{I}$ th global model $\mathcal{M}^{\mathcal{I}}$, the $\mathcal{I}$ th iteration training set's group $Group^{\mathcal{I}}$, the privacy guarantee ϵ
Output: the $(\mathcal{I} + 1)$ th training set's group $Group^{\mathcal{I}+1}$

- 1: **for** $z_i = (x_i, y_i)$ in D^{ben} **do**
- 2: **if** $\mathcal{N}_{z_i} < 2$ **then**
- 3: Continue
- 4: **end if**
- 5: **if** $y_i \notin \text{Topk}(F_{\mathcal{M}^{\mathcal{I}}}(x_i))$ **then**
- 6: Randomly remove one sample from $Group^{\mathcal{I}}_{z_i}$
- 7: Randomly choice y'_i and y''_i that are not originally in $Group^{\mathcal{I}}_{z_i}$
- 8: Add (x_i, y'_i) and (x_i, y''_i) to $Group^{\mathcal{I}}_{z_i}$ to form $Group^{\mathcal{I}+1}_{z_i}$
- 9: **else**
- 10: Randomly remove two samples from $Group^{\mathcal{I}}_{z_i}$
- 11: $y_i^b = y_i$ with probability $\frac{e^{\epsilon'}}{e^{\epsilon'} + C - 1}$
- 12: Or $y_i^b \in C \setminus \{y_i\}$ with probability $\frac{1}{e^{\epsilon'} + C - 1}$
- 13: Add (x_i, y_i^b) to $Group^{\mathcal{I}}_{z_i}$ to form $Group^{\mathcal{I}+1}_{z_i}$
- 14: **end if**
- 15: **end for**
- 16: **return** $Group^{\mathcal{I}+1}$ and train on $Group^{\mathcal{I}+1}$

and true labels, not on the other labels. Therefore, RR-Label is difficult to ensure the privacy of FL clients under TEPE.

C. Design of RR-Group

To defend against our proposed attack, we propose a membership inference defense method named Random Response with Group (RR-Group), which creates a group for each training sample, unconstrained by poisoning strategies in black-box settings. We specifically illustrate the algorithm steps of RR-Group in Algorithm 2. Each initialized group is composed of $\mathcal{N}_{z_i}$ samples, denoted as

$$Group^{\mathcal{I}}_{z_i} = \{z_i, z_i, \dots, z_i\}. \quad (16)$$

For each sample z_i , the data portion remains invariant, while the label portion is subject to dynamic adjustment via RR-Group. The most significant distinction between RR-Group and RR-Label lies in the fact that RR-Label invariably randomly alters the label of a sample to a single label with a specific probability, but RR-Group has the possibility of random labels regardless of the model prediction results. In addition, the defensive strategy adopted by RR-Label requires that the model predictions come from C_1 classes. However, prior to the convergence of the FL global model, prediction results may not guide the label selection process, which may undermine the practicality of the model. As a result, we construct respective groups for each training sample and dynamically adjust $\mathcal{N}_{z_i}$. As soon as the model makes an incorrect prediction, a new label is added to $Group^{\mathcal{I}}_{z_i}$. If the model's predictions are correct, it can be supposed that the sample is not under attack and the size of $Group^{\mathcal{I}}_{z_i}$ gradually

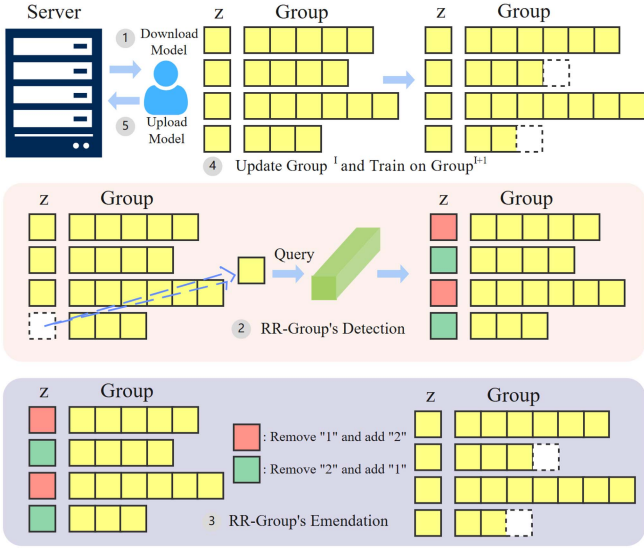


Fig. 4. RR-Group's workflow. Benign clients make dynamic adjustments to Groups via detection and emendation in every FL iteration based on the model from the preceding round. The green and red groups represent $Group_{z_i}^{trust}$ and $Group_{z_i}^{susp}$, respectively. "1" and "2" denote one sample and two samples within $Group_{z_i}$ respectively.

decreases. Therefore, Regardless of the model's predictions, RR-Group can reduce the leakage of membership information.

1) *RR-Group's Workflow*: Fig. 4 presents the RR-Group workflow of benign clients. First, a benign client downloads the global model from the server. Second, the client employs the global model to detect all of its training samples to divide $Groups^I$ into suspicious ones and trusted ones. Third, $Groups^I$'s Emendation enhances privacy protection. Finally, the client utilizes $Groups^{I+1}$ to train the global model and uploads the model to the server.

2) *Defense Analysis of RR-Group*: We define the trusted group and the suspicious group in a client's training set as $Group_{z_i}^{trust}$ and $Group_{z_i}^{susp}$ respectively, and assume that $Group_{z_i}^{susp}$ is affected by malicious clients' poisoning membership inference attacks, while $Group_{z_i}^{trust}$ is not.

$Group_{z_i}^{susp}$ is represented as

$$Group_{z_i}^{susp} = \{(x_i, y_i), (x_i, y_i), \dots, (x_i, y'), (x_i, y'')\}$$

$$\text{s.t. } \mathcal{L}_{\tilde{\mathcal{M}}^I(x_i, y_i)} - \mathcal{L}_{\tilde{\mathcal{M}}^I(x_i, y')} < 0$$

$$\mathcal{L}_{\tilde{\mathcal{M}}^I(x_i, y_i)} - \mathcal{L}_{\tilde{\mathcal{M}}^I(x_i, y'')} < 0, \quad (17)$$

where $y' \neq y_i$, $y'' \neq y_i$, and $\tilde{\mathcal{M}}^I$ is a model trained by $Group_{z_i}^{susp}$ as follows:

$$\tilde{\mathcal{M}}^I = \tilde{\mathcal{M}}^{I-1} - \lambda \frac{\partial \mathcal{L}_{\tilde{\mathcal{M}}^{I-1}}(Group_{z_i}^{susp})}{\partial \tilde{\mathcal{M}}^{I-1}}, \quad (18)$$

where λ represents the local training learning rate.

As the poisoning attack of malicious clients takes effect on non-defense, without loss of generality, we have

$$\mathcal{L}_{\mathcal{M}^I}(x_i, y_i) > \xi, \quad (19)$$

where the constant ξ represents the loss threshold of sample z_i on the model.

(19) demonstrates that as the poisoning continuously impacts the model, the global model experiences a model loss regarding the sample, which is larger than the threshold value ξ . Since we add $Group_{z_i}^{susp}$ to the model training in (18) (other labels help to weaken the performance of the target sample against poisoning attacks), which results in

$$\mathcal{L}_{\tilde{\mathcal{M}}^{I+1}}(x_i, y_i) - \mathcal{L}_{\mathcal{M}^I}(x_i, y_i) \geq 0. \quad (20)$$

Combining (19) and (20), and we have

$$\mathcal{L}_{\tilde{\mathcal{M}}^{I+1}}(x_i, y_i) > \xi. \quad (21)$$

This indicates that during the training process, the model performance of the sample z_i may not be better than the base case. Therefore, the prediction result $F_{\tilde{\mathcal{M}}^I}(x_i)$ of the model trained by a benign client under $Group_{z_i}^{susp}$ tends to be $p(\iota = \hat{\iota}) = \frac{1}{C}$, ($\hat{\iota} = 0, 1, \dots, C-1$). This means that the sample's poisoning resistance is weakened. Therefore, the difference between the member and non-member samples regarding the influence of poisoning is blurred. This ideal situation restricts all malicious updates that help distinguish the membership information of samples.

For $Group_{z_i}^{trust}$, RR-Group continuously reduces its samples, thereby reducing the risk of leakage of the sample's membership information. RR-Group makes member samples perform the same as non-member samples in terms of resistance to the influence of poisoning.

Consequently, it is arduous for malicious clients to conduct effective targeted poisoning attacks on target samples, and the change in (8) of TEPE observed by malicious clients also tends to be consistent over time.

3) *Differential Privacy Analysis*: At the same time, we consider the idea of DP to ensure that RR-Group satisfies the privacy guarantee of ϵ and $\epsilon' = \frac{\epsilon}{N_{z_i}}$ in Algorithm 2. Considering that any two inputs y and y' belong to C categories, for any possible output $\tilde{y}$, $\Pr[\text{RR-Group}(y) \text{ all is } \tilde{y}]$ is maximized when $y = \tilde{y}$. However, $\Pr[\text{RR-Group}(y) \text{ all is } \tilde{y}]$ is minimized when $y' \in C \setminus \{\tilde{y}\}$, and we have

$$\frac{\Pr[\text{RR-Group}(y) \text{ all is } \tilde{y}]}{\Pr[\text{RR-Group}(y') \text{ all is } \tilde{y}]} \leq \frac{(\frac{e^{\epsilon'}}{e^{\epsilon'} + C - 1})^{N_{z_i}}}{(\frac{1}{e^{\epsilon'} + C - 1})^{N_{z_i}}}$$

$$= \frac{e^{\epsilon}}{e^{\epsilon} + C - 1} = e^{\epsilon}. \quad (22)$$

Motivated by RR-Topk [52], RR-Group is equivalent to performing multiple rounds of the RR-Topk randomized response process on a single sample while dynamically adjusting the number of samples within the group. RR-Group dynamically adjusts the group size N_{z_i} , specifically expanding the group (by introducing more random labels) when the model makes an incorrect prediction. This strategy effectively establishes a "noisier communication channel" between y and y_p . Compared to the static RR-Label, RR-Group adaptively increases the upper bound precisely when the model is most vulnerable and susceptible to poisoning attacks (i.e., during prediction errors),

TABLE III
THE ARCHITECTURES OF FEDERATED MODELS

Dataset	Model Architecture
CIFAR-10	2 Conv., 2 Pool., and 3 FC layers
MNIST	2 Conv., 2 Pool., and 3 FC layers
Fashion-MNIST	2 Conv., 2 Pool., and 3 FC layers
Purchase-100	5 FC layers
CIFAR-100	2 Conv., 2 Pool., and 3 FC layers

thereby further reducing the maximum potential membership information contained within a sample.

By changing $Group_{z_i}$, we successfully solve the problem of balancing model performance and privacy protection caused by targeted poisoning attacks. Furthermore, while dynamically adjusting the group size, RR-Group also manages to reduce the computational burden. Consistent with Zheng et al. [24], the computational overhead introduced by RR-Group is limited.

VI. PERFORMANCE EVALUATION

A. Experimental Settings

1) *Datasets*: We utilize five datasets to evaluate our methodology, including Purchase-100 [73], MNIST [74], Fashion-MNIST [75], CIFAR-10, and CIFAR-100 [76]. Both CIFAR-10 and CIFAR-100 consist of 60,000 color images, each with a resolution of 32×32 pixels. CIFAR-10 comprises 10 classes, with 5,000 training images and 1,000 testing images per class. CIFAR-100 consists of 100 image categories, with 500 training images and 100 testing images per category. Purchase-100 encompasses shopping records from thousands of online customers. For our experiments, we utilize a simplified version of it. The entire set comprises 600 unique products, and for each product, users have a binary indicator indicating whether they purchased it or not, totaling 197,324 records. MNIST and Fashion-MNIST are both collections of 70,000 images, and each image is 28×28 pixels in size. The labels are one-dimensional arrays of length 10, representing the specific class of the image.

2) *Target Models*: The aggregation algorithm for the FL model employs FedAvg [77]. We set five clients, and distribute the dataset randomly to the clients. Each client performs five local iterations, while the FL global aggregation count is set to twenty-five. During each FL training iteration, the clients train the received FL model in a black-box manner using their datasets. For different datasets, we utilize distinct models, with the model structures outlined in Table III. Specifically, FC layer represents the fully connected layer in a Deep Neural Network (DNN) model. Conv. layer denotes convolutional layers, and Pool. layer signifies the max pooling layer in a Convolutional Neural Network (CNN) model.

3) *Settings of Attacks*: In the Spy's setting, the number of poisoned samples is set to 0.2%. To obtain the adversarial samples, we utilize an Adam [67] optimizer with a learning rate of 0.01 to optimize initial samples over 1,000 iterations. This optimization process targets a specific objective function to craft the poisoned samples. In the Adv's setting, the samples with poisoned labels are replicated 0.16%. This multiplication

TABLE IV
COMPARISON OF ATTACK METRICS ON DIFFERENT DATASETS

Attack	Metric	CIFAR-10	MNIST	F-MNIST	CIFAR-100	P-100
TEPE (Adv&Spy)	Accuracy	0.851	0.763	0.752	0.854	0.731
	Precision	0.818	0.731	0.725	0.709	0.672
	Recall	0.989	<u>0.761</u>	0.890	<u>0.975</u>	0.807
	F1-score	0.872	<u>0.745</u>	0.799	0.823	0.733
TEPE (Adv)	Accuracy	0.849	0.766	0.744	<u>0.845</u>	<u>0.730</u>
	Precision	0.817	0.730	0.719	0.706	0.668
	Recall	0.980	0.761	<u>0.882</u>	0.970	<u>0.805</u>
	F1-score	<u>0.869</u>	0.744	0.780	0.819	0.730
TEPE (Spy)	Accuracy	0.832	0.760	0.743	0.843	0.728
	Precision	0.804	0.727	0.717	0.704	0.668
	Recall	0.977	0.758	0.880	0.967	0.804
	F1-score	0.858	0.736	<u>0.779</u>	0.811	0.729
White-Box [17]	Accuracy	0.602	0.512	0.560	0.607	0.671
	Precision	0.670	0.519	0.553	0.582	0.643
	Recall	0.408	0.612	0.590	0.711	0.650
	F1-score	0.531	0.557	0.561	0.644	0.602
CMI [24]	Accuracy	0.707	0.697	0.570	0.832	0.730
	Precision	0.712	0.641	0.610	0.858	0.786
	Recall	0.708	0.750	0.641	0.845	0.721
	F1-score	0.718	0.707	0.619	0.865	<u>0.732</u>
TEAR [27]	Accuracy	0.829	0.738	0.736	0.786	0.702
	Precision	0.833	0.720	0.727	0.807	0.727
	Recall	0.847	0.749	0.719	0.844	0.753
	F1-score	0.842	0.730	0.723	<u>0.824</u>	0.728
L-Leaks [30]	Accuracy	0.850	0.765	0.751	0.830	0.707
	Precision	0.769	0.742	0.753	0.758	0.711
	Recall	<u>0.987</u>	0.760	0.811	0.989	0.744
	F1-score	0.869	0.749	0.776	<u>0.864</u>	0.725
Aster [78]	Accuracy	0.657	0.667	0.567	0.608	0.665
	Precision	0.612	0.633	0.533	0.558	0.686
	Recall	0.758	0.762	0.637	0.762	0.678
	F1-score	0.678	0.697	0.584	0.652	0.682

strategy enhances the Adv's ability to manipulate the model's behavior. We configure K-means [71] with a cluster number of 2 and an iteration count of 1,000.

4) *Settings of Defense*: In our settings, we use MemGuard [50], RR-Label [24] and DP-SGD [56] as comparison experiments to evaluate the defense capability of RR-Group against TEPE (Adv&Spy). For DP-SGD, we set the privacy budget $\epsilon = 5$ and $\delta = 10^{-5}$. For RR-Label, we adopt $q = 0.2$ in $Bernoulli(q)$, following the original RR-Label settings. For RR-Group, we initially set the size of Groups to 3 and adopt Top6 in Line 5 of Algorithm 2.

B. Evaluation of TEPE

In addition to the prior knowledge differences, the experiments are conducted under the same settings to compare the performance of TEPE with that of other MIAs. We use five datasets and five MIAs, including White-Box [17], CMI [24], TEAR [27], L-Leaks [30], and Aster [78], to conduct the comparative experiments.

1) *Performance of TEPE*: As demonstrated in Table IV, TEPE exhibits superior membership inference performance compared to four baseline attacks without requiring any knowledge of target models or benign clients' training data. All experimental results are presented as the mean of five independent runs, with standard deviations less than 0.01. On CIFAR-10, TEPE (Adv&Spy) achieves 85.1% attack accuracy, outperforming White-Box by 24.9 percentage points (pp), CMI

TABLE V
TRAINING AND TESTING ACCURACY COMPARISON WITHOUT AND WITH
POISONING

Strategy	Metric	CIFAR-10	MNIST	F-MNIST	CIFAR-100	P-100
Non-poison	Train Acc.	0.9480	1.0000	0.9936	0.7683	0.9912
	Test Acc.	0.7432	0.9661	0.7627	0.5044	0.4875
Adv&Spy	Train Acc.	0.9350	1.0000	0.9941	0.7677	0.9967
	Test Acc.	0.7377	0.9646	0.7490	0.5018	0.4751
Adv	Train Acc.	0.9351	1.0000	0.9933	0.7677	0.9967
	Test Acc.	0.7376	0.9655	0.7492	0.5037	0.4755
Spy	Train Acc.	0.9476	1.0000	0.9941	0.7678	0.9966
	Test Acc.	0.7428	0.9659	0.7494	0.5042	0.4760

by 14.4 pp and TEAR by 2.2 pp, while maintaining a 0.872 F1-score that surpasses all baselines by 0.3-34.1 pp. For MNIST, TEPE (Adv&Spy) shows comparable performance (76.3% accuracy, 0.745 F1-score) to the strongest baseline (L-Leaks), with only 0.2 pp and 0.4 pp differences, respectively. The advantage becomes more pronounced on Fashion-MNIST, where TEPE (Adv&Spy)'s 0.799 F1-score leads all competitors. TEPE (Adv&Spy) achieves peak performance on CIFAR-100 with 85.4% accuracy and 0.823 F1-score, though with 9.8 pp lower precision than TEAR and with 4.2 pp lower F1-score than CMI. Finally, on Purchase-100, TEPE (Adv&Spy) dominates all metrics except precision, demonstrating robust attack capability across diverse datasets. TEPE (Adv) and TEPE (Spy) achieve suboptimal results on certain metrics, demonstrating that the synergistic effect of the Adv&Spy strategy more effectively leaks membership privacy. The simultaneous distortion of decision boundaries around target samples leads to an amplified disparity between member and non-member samples. Although Aster also requires no prior knowledge, its prediction sensitivity scheme may not apply to FL, leading to poor performance. These results collectively establish TEPE as a potent threat to federated learning privacy.

To evaluate the impacts of the poisoning strategy on model performance, we conduct experiments on five different datasets. The results, as shown in Table V, indicate that our poisoning strategy has a negligible effect on the model's performance, thus demonstrating the stealthiness of our attack.

Overall, TEPE demonstrates strong inference capabilities without any prior knowledge of the model and benign clients' training data. One possible reason is that quantifying poisoning effects is more adept at mining the membership information of target samples embedded within the target model. Even though some indicators are slightly lower than those of other comparative experiments, our method excels in requiring fewer conditions for inference. Through minor sacrifices, we achieve a more practical attack algorithm that does not require any prior knowledge of the model and benign clients' training data. All the subsequent ablation experiments are based on TEPE (Adv&Spy).

2) *Impacts of the Poisoned Label Selection:* Poisoned label selection strategies of the poisoning attack have the difference in the $Mentr_{PE}$ between the member and non-member samples

during the FL training iterations. Four distinct strategies are considered:

- *Suboptimal:* This means that instead of the true top-1 predicted class, the poisoning attack selects the class that ranks second in the prediction results.
- *Worst:* This means that the poisoning attack chooses the class that ranks the lowest in the prediction results.
- *Random-alone:* In this strategy, a random class is chosen as the poisoned label, without considering any specific ranking or prediction performance.
- *Random-multi:* This strategy involves randomly selecting multiple classes to use as poisoned labels, potentially increasing the uncertainty and confusion for the model.

Fig. 5 indicates that under a relatively small number of iterations, the gap between the member and non-member samples for Random-multi is quite small, which is attributed to the fact that multiple class samples may mislead the model during the initial stages of training. The other three strategies have apparent gaps between the member and non-member samples. Therefore, there is no need to specifically designate the choice of the poisoned label. As shown in Fig. 6, {2, 3, 4, 5} all can affect the decision boundary of {7}. Just make sure the poisoned labels are as uniform as possible.

3) *Impacts of the Poisoned Samples Proportion:* In TEPE, the proportion of the Adv's poisoned samples affects the performance of the inference attack. We conduct multiple experiments by increasing the proportion of poisoned samples from 0.02% to 0.24% in Fig. 7. For CIFAR-10 and Purchase-100, the inference performance improves as the Adv's poisoned samples increase. For example, when 0.2% poisoned samples are injected into CIFAR-10, TEPE achieves a recall of 0.982 and a precision of 0.779. However, with merely a single replication of poisoned samples, both recall and precision drop substantially to 0.6 and 0.533, respectively. Too few poisoned samples may not be able to achieve the poisoning effect, resulting in poor inference performance.

In TEPE, the proportion of the Spy's poisoned samples affects the effectiveness of inference attacks. We increase the proportion of poisoned samples from 0.1% to 1%. Fig. 8 illustrates the effect of the proportion of Spy's poisoned samples on inference attacks. On CIFAR-10 and Purchase-100, too many or too few Spy's poisoned samples can degrade the attack performance. Based on the analysis of the experimental results, the poisoned samples are insufficient to disturb the decision boundary of the target samples significantly. However, if there are too many Spy's poisoned samples, the membership information of the target sample may be shielded, thus reducing the attack performance.

Therefore, an appropriate proportion of poisoning samples has an impact on the attack effect. For malicious clients, employing a larger proportion of poisoned samples is advantageous, as the inference attack performance eventually converges to its optimal level. In contrast, for spies, the proportion of poisoned samples must be carefully calibrated, neither excessive nor insufficient. Therefore, the limitation of TEPE is that the performance of inference attacks may be affected by the quality and proportion of poisoned samples. We believe that the attack

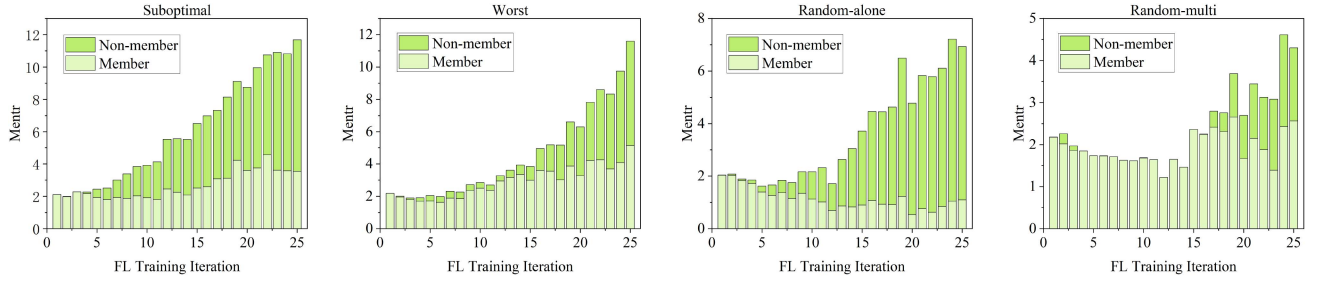


Fig. 5. A visual presentation on $Mentr_{PE}$ comparing four mislabelling strategies using TEPE on CIFAR-10. Different label choices can cause $Mentr_{PE}$ to behave differently between the member and non-member samples. In our setup, apart from the Random-multi, the other three mislabelling strategies can be well distinguished.

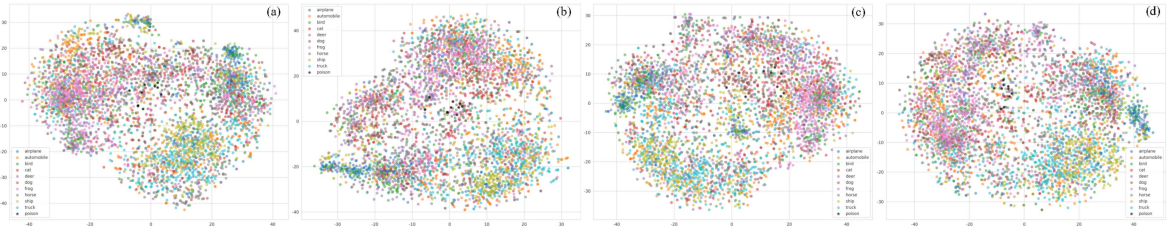


Fig. 6. Visualization of CIFAR-10 representations using t-SNE [79]. The dark stars denote the poisoned samples on the Spy strategy. The truth label is 7 (frog). (a) The poisoned label is 2 (bird). (b) The poisoned label is 3 (cat). (c) The poisoned label is 4 (deer). (d) The poisoned label is 5 (dog). To ensure visual clarity, each figure displays only a limited number of poisoned samples. These poisoned samples induce a shift in the decision boundaries of target samples, thereby highlighting the disparity between member and non-member samples, which is manifested in (8) and Fig. 3.

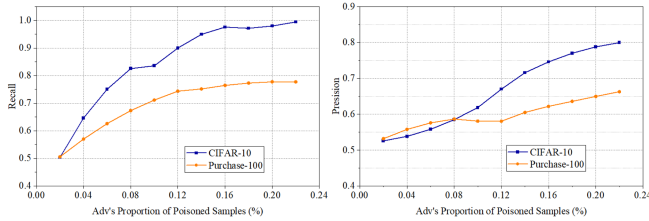


Fig. 7. Precision and recall of TEPE vs. proportion of Adv's poisoned samples.

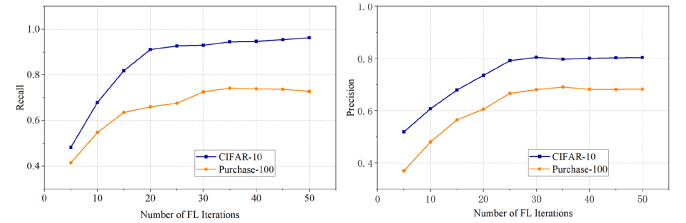


Fig. 9. Precision and recall of TEPE vs. number of FL iterations.

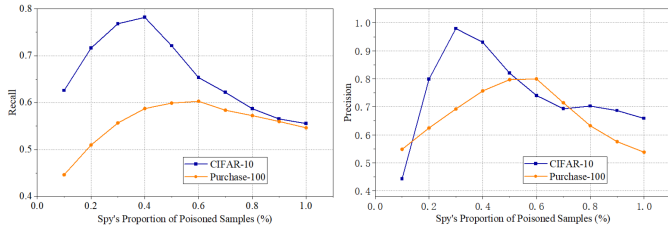


Fig. 8. Precision and recall of TEPE vs. the proportion of Spy's poisoned samples.

principle in TEPE can be applied in any black-box FL settings, but some tweaks may be necessary.

4) *Impacts of the Iteration Number*: The iteration number refers to the aggregation frequency of the global model. As mentioned earlier, through repeated experiments, we continually increase the iteration numbers on both Purchase-100 and CIFAR-10 to calculate the inference attack performance.

Fig. 9 indicates that as the iteration number increases, better inference attack performance can be achieved. For CIFAR-10, when comparing the performance between 10 iterations and 50 iterations, the precision increases from 0.521 to 0.802, and the recall rises from 0.64 to 0.978. As the iteration number continues to grow, the attack performance first improves and then stabilizes.

Therefore, a large iteration number is conducive to the leakage of membership information. A little iteration number leads to insufficient poisoning effects from the poisoned samples, resulting in larger perturbations in quantifying the temporal evolution of poisoning effects, thereby reducing the inference attack performance.

5) *Impacts of the Proportion of Malicious Clients*: Table VII shows the impacts of the proportion of malicious clients among all FL clients. For CIFAR-10, the attack accuracy peaks at 0.853 with the 15% proportion, followed by a gradual decline as the proportion increases. In contrast, Purchase-100 exhibits

TABLE VI
TEPE'S F1-SCORE UNDER THE ROBUST AGGREGATION RULES

Aggregation Rules	CIFAR-10	MNIST	CIFAR-100	P-100
FedAvg [77]	0.851	0.763	0.854	0.731
Krum [80]	0.851	0.763	0.854	0.731
Trimmed Mean [80]	0.851	0.763	0.854	0.731

TABLE VII
TEPE'S ACCURACY VS. THE PROPORTION OF MALICIOUS CLIENTS

Proportion	10%	15%	20%	25%	30%	35%
CIFAR-10 Mem. Acc.	0.797	0.853	0.851	0.819	0.804	0.793
Purchase-100 Mem. Acc.	0.724	0.725	0.731	0.739	0.727	0.711

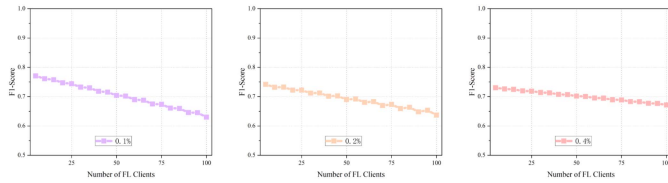


Fig. 10. F1-score of TEPE vs. number of FL clients. The proportion of poisoned samples is set to low (0.1%), medium (0.2%), and high (0.4%) to evaluate the impacts of the number of FL clients.

greater stability, with accuracy fluctuating narrowly between 0.724 and 0.739. The highest accuracy (0.739) is achieved when the proportion of malicious clients reaches 25%.

Therefore, malicious clients can collude to manipulate the global models while strategically evading anomaly detection mechanisms, thereby achieving superior attack performance. Malicious clients can insidiously degrade the model's privacy over time by gradually reducing the number of poisoned samples, where each individual sample has a barely perceptible yet cumulative effect.

6) *Impacts of the Number of FL Clients*: Fig. 10 shows the impacts of the number of FL clients. We set the proportion of poisoned samples to low (0.1%), medium (0.2%), and high (0.4%) to discuss the impact of the number of clients. Although an increase in the number of clients leads to a slight degradation in inference performance, this effect is mitigated at higher poisoning proportions (e.g., 0.4%), where the attack performance becomes less sensitive to the number of FL clients.

Therefore, the robustness of TEPE against the number of FL clients can be enhanced by increasing the proportion of poisoned samples.

7) *Impacts of the Aggregation Rules*: To validate the robustness and generalizability of the proposed method across different aggregation rules, we evaluate TEPE under various Byzantine-robust aggregation mechanisms. The results are summarized in Table VI. We conduct experiments using three mainstream aggregation rules—FedAvg [77], Krum, and Trimmed Mean [80]—and reported their F1-scores. As shown in the table, the F1-scores of TEPE remain stable across all datasets, regardless of the aggregation rule, with no significant performance degradation. For instance, on CIFAR-10, the F1-scores under

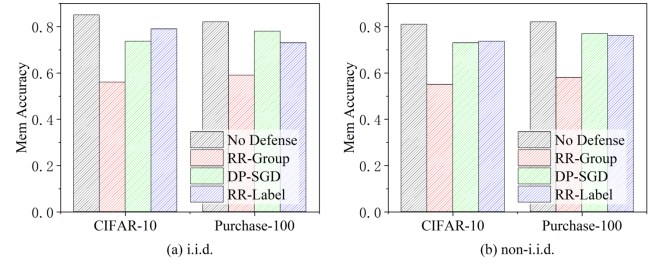


Fig. 11. Compare the average membership inference accuracy of TEPE over three runs under different methods of defense. In most cases, RR-Group achieves better defense performance under the i.i.d. and non-i.i.d. settings.

FedAvg, Krum, and Trimmed Mean are all 0.851. This indicates that our attack strategy does not rely on any specific aggregation mechanism and remains effective even under strongly robust aggregation rules such as Krum and Trimmed Mean, without being easily detected.

Therefore, the experimental results strongly demonstrate that the TEPE attack exhibits low sensitivity to aggregation rules and maintains its efficacy across a range of robust aggregation rules. This further implies that existing aggregation strategies struggle to defend against such stealthy attacks aimed at privacy leakage.

C. Experiments of RR-Group

In our experiments, we evaluate the defensive efficacy of RR-Group against TEPE by comparing it with three state-of-the-art defense mechanisms: MemGuard [50], RR-Label [24], and DP-SGD [56]. This comparative analysis assesses the performance of each approach in mitigating membership inference risks under targeted poisoning attacks.

1) *Performance of RR-Group*: Fig. 11 shows the main experimental defense results. As mentioned in Section V-B, DP-SGD does have a certain impact on the attack performance of TEPE. However, even if test accuracy is sacrificed, DP-SGD still struggles to achieve satisfactory defense performance. Specifically, on CIFAR-10 and Purchase-100, the membership inference accuracy on DP-SGD reaches above 0.7. We analyze that DP-SGD can affect the model training process to a certain extent. However, the strong gradient disturbance caused by poisoning is much larger than that caused by noise, which indicates that TEPE has a certain stability. We also note that RR-Label is difficult to defend against TEPE. To prove the reliability of our algorithm, we remove 10% of the training set to meet the i.i.d. setting of RR-Label. Fig. 11(a) shows that in the i.i.d. setting, RR-Group's membership inference accuracy reaches 0.56 and 0.59 on CIFAR-10 and Purchase-100 respectively. Fig. 11(b) shows the comparison results of non-i.i.d. settings on different defense methods. Compared with the other two methods, RR-Group can achieve excellent defense effects and its membership inference accuracy reaches 0.55 and 0.58 on CIFAR-10 and Purchase-100 respectively. It should be noted that RR-Label makes it difficult to play an effective defense against TEPE. The possible reason is that, as mentioned in Section V-B, RR-Label can randomly

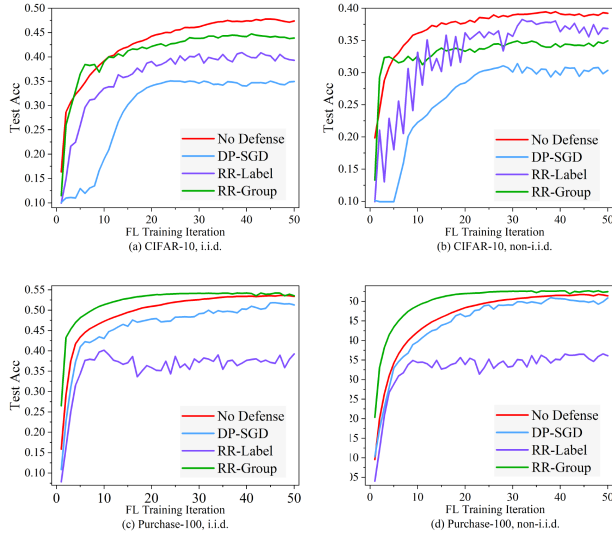


Fig. 12. Test accuracy comparison of four defense mechanisms—No Defense, DP-SGD, RR-Label, and RR-Group—under both i.i.d. and non-i.i.d. data distributions across two benchmark datasets: CIFAR-10 and Purchase-100.

TABLE VIII
TEPE'S F1-SCORE UNDER THE DEFENSE OF MEMGUARD

Defense	CIFAR-10	MNIST	CIFAR-100	P-100
No Defense	0.851	0.763	0.854	0.731
Memguard ($\epsilon = 0.5$)	0.851	0.763	0.854	0.731
Memguard ($\epsilon = 1$)	0.851	0.763	0.854	0.731

interfere with the labels of some member samples, but it ignores the poisoning ability of malicious clients.

Fig. 12 shows the impacts of different defense methods on model performance. We find that RR-Group achieves competitive model performance on both i.i.d. and non-i.i.d. settings. The test accuracy of DP-SGD is much higher than that of Zheng et al. [24]. This is attributed to the limited number of FL clients, while DP-SGD maintains satisfactory test accuracy. In addition, we also find that RR-Group can achieve greater test accuracy in fewer FL training iterations than the other three methods, possibly because RR-Group has $\mathcal{N}_{z_i}$ copies of each data at the beginning of training, resulting in greater model performance improvements. Interestingly, RR-Group achieved better test accuracy on Purchase-100 than No Defense, with 0.136 pp and 1.077 pp higher test accuracy in the i.i.d. and non-i.i.d. settings, respectively, while this is not the case with CIFAR-10. The performance difference between CIFAR-10 and Purchase-100 is reflected in the fact that the number of classes in Purchase-100 is much larger than CIFAR-10, and RR-Group can play a better role. In the non-i.i.d. setting of CIFAR-10, the test accuracy of RR-Label is greater than that of RR-Group. As mentioned in Section V-B, the possible reason is that our poisoning strategy leads to $RR - Label\{(x, y)\} = (x, y)$. Table VIII shows that MemGuard, the representative of inference-stage defenses ignores the connection of prediction confidence during the model

TABLE IX
MODEL PERFORMANCE AND DEFENSE PERFORMANCE VS. $Topk$ UNDER CIFAR-10 AND THE I.I.D. SETTINGS

$Topk$	1	2	3	4	5	6	7
Test Acc.	0.3084	0.3116	0.3446	0.3501	0.3832	0.3969	0.3970
Mem. Acc.	0.551	0.582	0.595	0.597	0.574	0.542	0.542

TABLE X
MODEL PERFORMANCE AND DEFENSE PERFORMANCE VS. $\mathcal{N}_{z_i}$ AT INITIALIZATION UNDER PURCHASE-100 AND THE I.I.D. SETTINGS

$\mathcal{N}_{z_i}$	3	5	7	9	11	13
Test Acc.	0.5406	0.5424	0.5461	0.5573	0.5489	0.5589
Mem. Acc.	0.543	0.553	0.746	0.770	0.774	0.841

training process. TEPE can observe the temporal evolution of poisoning effects to escape the protection of MemGuard.

In conclusion, we confirm that RR-Group is an effective defense method against membership inference attacks that exploit poisoning effects. On the premise of ensuring certain model performance, RR-Group makes the performance index of membership inference attacks close to random guessing, achieving a balance between model performance and privacy.

2) *Impacts of $Topk$ Selection:* Table IX shows that the smaller k , the lower the test accuracy. The reason is that a smaller k indicates more complex labels within the sample's *Groups*. Through FL training iterations, the excessive poisoned labels in the sample groups degrade model performance. The attack accuracy generally approximates random guessing but does not follow a monotonic trend. This can be analyzed as follows: When k is relatively small, RR-Group applies the random process in Lines 6-8 of Algorithm 2 to more training samples, sacrificing model performance to achieve better defense capability. When k is relatively large, RR-Group applies the Random Response process in Lines 10-13 of Algorithm 2 to more training samples, maintaining model performance while reducing the accuracy of MIAs.

Therefore, the selection of $Topk$ has a limited influence on defense performance close to random guessing but has a certain influence on model performance. A larger k has a slight improvement on the model performance (Test Acc.).

3) *Impacts of Initial Group's Size:* Table X demonstrates the impacts of the initial *Group* size $\mathcal{N}_{z_i}$ in the RR-Group on both defense and model performance. With the increase of $\mathcal{N}_{z_i}$, the model's membership inference defense performance gradually declines, while model performance improves slightly. The presence of excessive samples in the initial *Group* enhances the discernibility of the target sample's membership information, thereby aggravating client-level membership privacy leakage. However, the number of samples in each *Group* is quickly modified by RR-Group, so the initial *Group* size $\mathcal{N}_{z_i}$ has little impact on model performance.

Therefore, *Group* size $\mathcal{N}_{z_i}$ should not be set too large, as a smaller $\mathcal{N}_{z_i}$ can achieve both strong model performance and

effective privacy protection. An overly large $\mathcal{N}_{z_i}$ (>7) actually increases the risk of members' privacy leakage.

VII. CONCLUSION

In this paper, we have innovatively proposed the Temporal Evolution of the Poisoning Effects (TEPE), marking the MIAs specifically designed for Federated Learning (FL) models. TEPE leverages the unique temporal evolution patterns of target samples after being subjected to our proposed poisoning attacks (Adv, Spy, Adv&Spy), extracting crucial membership features to infer membership privacy. Through extensive and rigorous experiments across a diverse range of FL models and real-world datasets, we show that TEPE demonstrates superior performance compared to the current state-of-the-art MIA methods. Critically, TEPE does not require any prior knowledge about the target models or benign clients' training data, significantly broadening its application scenarios and threat potential. We have discussed two defenses of different stages (inference stage defenses and training stage defenses) and proposed a novel defense method, Random Response with Group (RR-Group), which addresses the balance between model performance and privacy protection under targeted poisoning attacks. Through continued research and efforts, we anticipate further strengthening the capability of FL to safeguard user data privacy, facilitating its safer and more reliable application across various industries. We firmly believe that this research not only deepens our understanding of the data privacy leakage risks faced by FL models in practical deployments but also provides invaluable insights into exploring more efficient and targeted defense strategies against such membership inference attacks.

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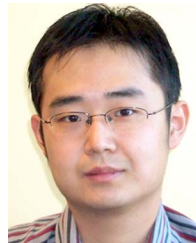
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